

Администрирование сетевых подсистем

Установка и настройка MariaDB (Лабораторная работа №6)

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Цели и задачи работы

Приобретение практических навыков по установке и конфигурированию СУБД MariaDB.

Выполнение лабораторной работы

Установка MariaDB

```
Installing      : mariadb-backup-3:10.11.11-1.el10.x86_64          7/11
Installing      : mariadb-gssapi-server-3:10.11.11-1.el10.x86_64  8/11
Installing      : mariadb-server-utils-3:10.11.11-1.el10.x86_64  9/11
Running scriptlet: mariadb-server-3:10.11.11-1.el10.x86_64      10/11
Installing      : mariadb-server-3:10.11.11-1.el10.x86_64      10/11
Running scriptlet: mariadb-server-3:10.11.11-1.el10.x86_64      10/11
Installing      : mariadb-3:10.11.11-1.el10.x86_64              11/11
Running scriptlet: mysql-selinux-1.0.14-1.el10_0.noarch          11/11
Running scriptlet: mariadb-3:10.11.11-1.el10.x86_64             11/11

Installed:
  mariadb-3:10.11.11-1.el10.x86_64
  mariadb-client-utils-3:10.11.11-1.el10.x86_64
  mariadb-errmsg-3:10.11.11-1.el10.noarch
  mariadb-server-3:10.11.11-1.el10.x86_64
  mysql-selinux-1.0.14-1.el10_0.noarch
  perl-Sys-Hostname-1.25-512.2.el10_0.x86_64
  mariadb-backup-3:10.11.11-1.el10.x86_64
  mariadb-common-3:10.11.11-1.el10.noarch
  mariadb-gssapi-server-3:10.11.11-1.el10.x86_64
  mariadb-server-utils-3:10.11.11-1.el10.x86_64
  perl-DBD-MariaDB-1.23-10.el10.x86_64

Complete!
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# ls /etc/my.cnf.d/
auth_gssapi.cnf  enable_encryption.preset  mysql-clients.cnf  provider_lz4.cnf  provider_snappy.cnf
client.cnf      mariadb-server.cnf        provider_bzip2.cnf  provider_lzo.cnf  spider.cnf
[root@server.manovichkov.net ~]# systemctl start mariadb
[root@server.manovichkov.net ~]# systemctl enable mariadb
Created symlink '/etc/systemd/system/mysqld.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/mysqld.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service' → '/usr/lib/systemd/system/mariadb.service'.
[root@server.manovichkov.net ~]# ss -tulpen | grep mysql
[root@server.manovichkov.net ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.
```

Настройка безопасности

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n]
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n]
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] █

Работа с интерактивной оболочкой

General information about MariaDB can be found at
<http://mariadb.org>

List of all client commands:

Note that all text commands must be first on line and end with ';'.

```
?      (\?) Synonym for 'help'.
charset (\C) Switch to another charset. Might be needed for processing binlog with multi-byte charsets.
clear   (\c) Clear the current input statement.
connect (\r) Reconnect to the server. Optional arguments are db and host.
delimiter (\d) Set statement delimiter.
edit    (\e) Edit command with $EDITOR.
ego     (\G) Send command to MariaDB server, display result vertically.
exit    (\q) Exit mysql. Same as quit.
go      (\g) Send command to MariaDB server.
help    (\h) Display this help.
nopager (\n) Disable pager, print to stdout.
notee   (\t) Don't write into outfile.
nowarning (\w) Don't show warnings after every statement.
pager   (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print   (\p) Print current command.
prompt  (\R) Change your mysql prompt.
quit    (\q) Quit mysql.
rehash  (\#) Rebuild completion hash.
sandbox (\-) Disallow commands that access the file system (except \P without an argument and \e).
source  (\.) Execute an SQL script file. Takes a file name as an argument.
status  (\s) Get status information from the server.
system  (\!) Execute a system shell command.
tee     (\T) Set outfile [to_outfile]. Append everything into given outfile.
use     (\u) Use another database. Takes database name as argument.
warnings (\W) Show warnings after every statement.
```

For server side help, type 'help contents'

MariaDB [(none)]>

```
MariaDB [(none)]> SHOW DATABASES;
```

Database
information_schema
mysql
performance_schema
sys

4 rows in set (0.000 sec)

Рис. 4: Просмотр системных баз данных

Конфигурация кодировки

```
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.11.11-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

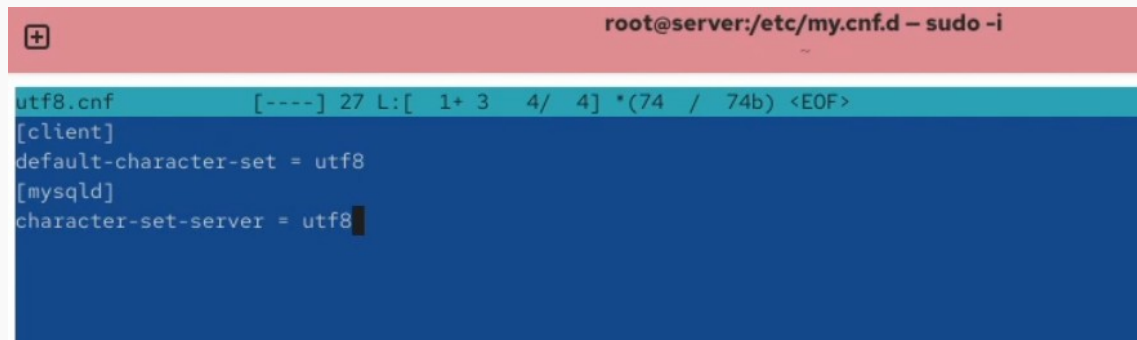
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.11-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          14
Current database:
Current user:           root@localhost
SSL:                   Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.11-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    latin1
Db      characterset:    latin1
Client characterset:    utf8mb3
Conn.  characterset:    utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 5 min 35 sec

Threads: 1  Questions: 27  Slow queries: 0  Opens: 20  Open tables: 13  Queries per second avg: 0.080
-----
MariaDB [(none)]>
```

Конфигурация кодировки



A terminal window with a red title bar. The title bar contains a plus icon on the left and the text "root@server:/etc/my.cnf.d – sudo -i" on the right. The terminal content shows the creation of a file named "utf8.cnf". The first line is a status bar: "utf8.cnf [----] 27 L:[1+ 3 4/ 4] *(74 / 74b) <EOF>". The subsequent lines are the file's contents: "[client]", "default-character-set = utf8", "[mysqld]", and "character-set-server = utf8". A black cursor is positioned at the end of the last line.

```
root@server:/etc/my.cnf.d – sudo -i

utf8.cnf [----] 27 L:[ 1+ 3 4/ 4] *(74 / 74b) <EOF>
[client]
default-character-set = utf8
[mysqld]
character-set-server = utf8
```

Рис. 6: Создание файла конфигурации utf8.cnf

Конфигурация кодировки

```
Your MariaDB connection id is 14
Server version: 10.11.11-MariaDB MariaDB Server
```

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
MariaDB [(none)]> status
```

```
-----
```

```
mysql Ver 15.1 Distrib 10.11.11-MariaDB, for Linux (x86_64) using EditLine wrapper
```

```
Connection id:          14
Current database:
Current user:            root@localhost
SSL:                     Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.11.11-MariaDB MariaDB Server
Protocol version:        10
Connection:              Localhost via UNIX socket
Server characterset:     latin1
Db characterset:         latin1
Client characterset:     utf8mb3
Conn. characterset:      utf8mb3
UNIX socket:             /var/lib/mysql/mysql.sock
Uptime:                  5 min 35 sec
```

```
Threads: 1 Questions: 27 Slow queries: 0 Opens: 20 Open tables: 13 Queries per second avg: 0.080
```

```
-----
```

```
MariaDB [(none)]> exit
```

Создание базы данных

```
MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.000 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.006 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов', 'Москва');
Query OK, 1 row affected (0.002 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров', 'Сочи');
Query OK, 1 row affected (0.001 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров', 'Дубна');
Query OK, 1 row affected (0.001 sec)

MariaDB [addressbook]> SELECT * FROM city;
+-----+-----+
| name      | city      |
+-----+-----+
| Иванов    | Москва    |
| Петров    | Сочи      |
| Сидоров    | Дубна     |
+-----+-----+
3 rows in set (0.000 sec)

MariaDB [addressbook]> █
```

Создание базы данных

```
3 rows in set (0.000 sec)

MariaDB [addressbook]>
MariaDB [addressbook]> CREATE USER manovichkov@%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,CREATE,INSERT,UPDATE,DELETE ON addressbook.* TO manovichkov@%';
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [addressbook]> DESCRIBE CITY;
ERROR 1146 (42502): Table 'addressbook.CITY' doesn't exist
MariaDB [addressbook]> DESCRIBE city;
+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+
2 rows in set (0.000 sec)

MariaDB [addressbook]> quit
Bye
[root@server.manovichkov.net my.cnf.d]# mysqlshow -u root -p
Enter password:
+-----+
| Databases |
+-----+
| addressbook |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
[root@server.manovichkov.net my.cnf.d]# mysqlshow -u manovichkov -p addressbook
Enter password:
█
```

Рис. 9: Создание пользователя и проверка базы данных

```
[root@server.manovichkov.net my.cnf.d]# mysqlshow -u manovichkov -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city   |
+-----+
[root@server.manovichkov.net my.cnf.d]#
[root@server.manovichkov.net my.cnf.d]# mkdir -p /var/backup
[root@server.manovichkov.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
```

Рис. 10: Проверка таблицы city в базе addressbook

```
[root@server.manovichkov.net my.cnf.d]#  
[root@server.manovichkov.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql  
Enter password:  
[root@server.manovichkov.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook  
Enter password:  
[root@server.manovichkov.net my.cnf.d]#  
[root@server.manovichkov.net my.cnf.d]# cd /vagrant/provision/server/  
[root@server.manovichkov.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d  
[root@server.manovichkov.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup  
[root@server.manovichkov.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/  
[root@server.manovichkov.net server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/  
[root@server.manovichkov.net server]#
```

Рис. 11: Создание и восстановление резервных копий базы addressbook

Выводы по проделанной работе

В ходе работы была установлена и сконфигурирована система управления базами данных MariaDB.

Реализованы настройка безопасности, создание базы данных и таблиц, работа с пользователями, а также процедуры резервного копирования и восстановления.